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Non-uniform truss modelling and energy consumption in adaptive space lattice manufacturing for steel structures

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ABSTRACT: The present paper studies the energetic footprint of Adaptive Space Lattice Manufacturing (ASLM), a new metallic-based AM process joining solid rods through laser spot welding. Associated energy costs are evaluated and compared with other metallic-based AM processes - machining, casting, powder-based AM and wire-based AM. Starting with measures of the energy consumption on site, a comprehensive study is performed across different units of measure (kWh/kg, kWh/rod, kWh/m3, kWh/I-beam). An energy breakdown is provided, including a material-process balance assessment through the integration of material embodied energy. Finally, the impact of design opportunities offered by ASLM is studied, evaluating various unit-cells lattice geometries across a sensitivity study. ASLM is found to achieve an 88,6-99,7% energy reduction compared to other AM technologies, and lattice configurations assessed feature 1:10 energy consumption differences.

1 INTRODUCTION

1.1 *Energy costs of metallic-based additive manufacturing*

Investigations into metallic-based additive manufacturing (AM) have been increasingly successful over the past decade in developing processes and applications for the AEC industry, from connectors and localized reinforcements to bridges or beams (Galjaard 2015, Camacho 2018, Costanzi 2022, Kyvelou 2022, Costanzi 2024). These fabrication processes are considered promising for a faster production of highly performing and custom components in the AEC industry (Buchanan 2019).

However, research has also shown the significant impact of such manufacturing processes on the environment, particularly their high energy consumption. While conventional manufacturing techniques for metals are already energy-demanding, digital fabrication processes increase the demand even further (Bekker 2018). Moreover, in AM processes the power can represent up to 80% of the impacts, in comparison to the materials and machine, making this aspect instrumental (Faludi 2017). Despite enabling a customization of parts that can play a key role in optimizing components (Galjaard 2015), the high energy demand requested by metallic AM processes questions their potential in architecture.

1.2 *Adaptive Space Lattice Manufacturing*

Aiming at proposing an alternative to 3D-printing processes, recent research in metallic-based manufacturing has explored other venues for the fabrication of non-standard, highly performing AEC components. Amongst others, Adaptive Space Lattice Manufacturing (ASLM) is a recent development in non-printing AM processes (Dirrenberger et al. 2024). Instead of

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melting and layering materials, it assembles solid rods in any direction using laser spot welding (Figure 1 and Figure 2). The robot places the rod at a predetermined location, the laser beam welds it at one end and cuts it at the other end. The cut is obtained by melting the cross-section of the rod at the cut point with the laser and pulling on it with a robot motion before it solidifies. ASLM appears as a promising metallic-based AM process given the preservation of material integrity it enables, and the material savings allowed by the truss geometries

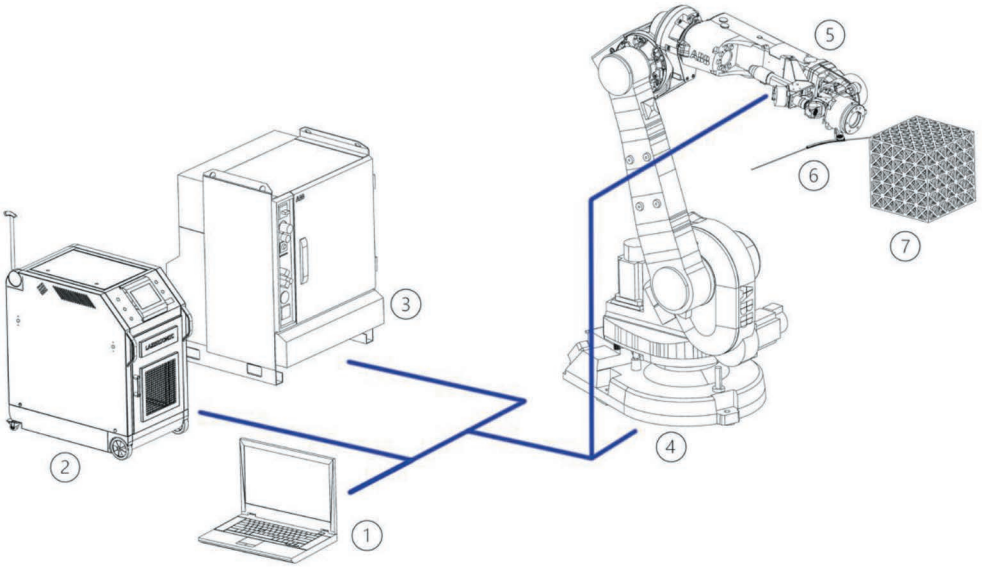


Figure 1. Schematic of the ASLM process: 1. System control; 2. Laser source; 3. Robot controller; 4. 6-axis robot; 5. Laser beam steering; 6. Wire holder; 7. ASLM-produced structure.

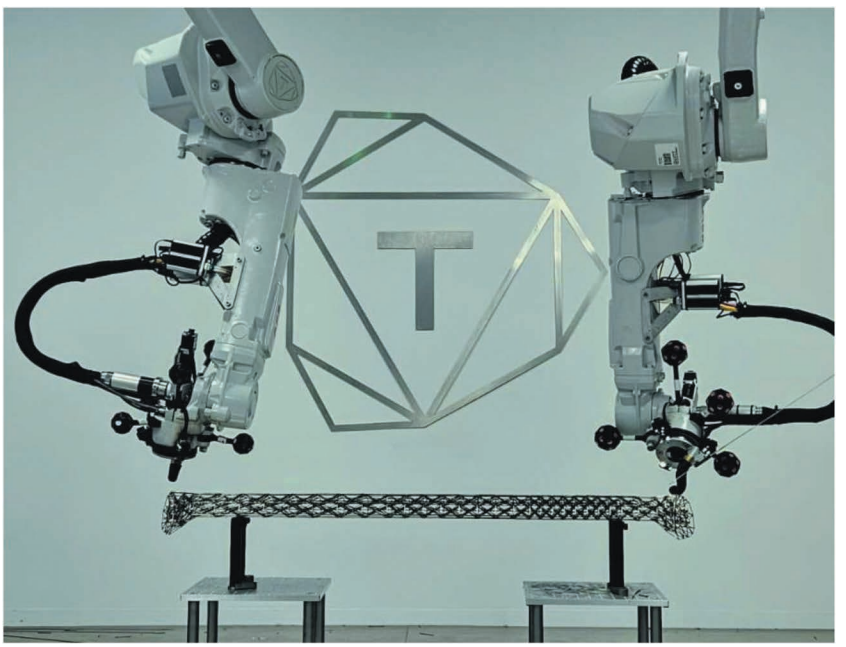


Figure 2. Picture of the ASLM process.

fabricated. Furthermore, as the energy consumption of the process is condensed in short spikes as the laser welds rods, it is assumed to be much lower than any other metallic-based AM process. The ASLM process presents promising application cases for AEC, producing truss members for various types of structures such as beams, or larger scale structural systems discretized into trusses. Performance studies have been conducted on ASLM trusses and show the predictable buckling behaviour under load of these structures (Dirrenberger et al. 2024). Feasibility studies demonstrate the ability of ASLM to mitigate common defects found in steel AM, such as surface roughness and porosity.

1.3 *Understanding the energy costs of ASLM*

Despite the promising use cases developed for the ASLM process, its energy bill has not been assessed yet. The present paper therefore studies the energetic footprint of ASLM, discussing its potential applications for construction and the associated energy costs that would ensue, in comparison with similarly performing metallic-based AM processes.

A comprehensive study of the energy consumption of ASLM is provided. Starting with measures on site, an evaluation is performed across different units of measure (kWh/kg, kWh/rod, kWh/m3, kWh/I-beam). A comparison is performed with machining, casting, powder-based AM and wire-based AM. An energy breakdown is provided, including a material-process balance assessment through the integration of material embodied energy. Finally, the impact of design opportunities offered by ASLM is studied, evaluating various unit-cells lattice geometries across a sensitivity study.

2 METHODOLOGY

2.1 *On-site energy consumption measurements*

As the focus of the research is to understand the energy footprint of ASLM, notably to compare it to other metal forming processes available on the market, a key aspect in methods selections is to enable the obtention of data usable in such comparison. Therefore, aligning with methods of measure adopted in other research regarding the energy footprint of metal forming processes and digital metal forming processes is the chosen starting point. A large part of the literature on the evaluation of energy consumption in an environmental perspective relies on on-site measures with a wattmeter. In particular (Bekker 2023) have chosen this method to evaluate the energy footprint of wire-arc AM, a major process to compare ASLM to. Furthermore, on-site wattmeter measures allow to examine the breakdown of the energy footprint between different parts of the system as well as the energetic cost of idle time periods. This precision in understanding the repartition of energy costs is instrumental in bettering the environmental impacts of the process and in leveraging environmental assessments to achieve energy savings. Measures have been executed at our partner Tetmet's workshop, in La Défense, France. The list of equipment used in the ASML process that are accounted for in the measures is detailed in section 3.2. Measures were conducted for 30 rod assembly cycles (3.6 sec each, 0.6s of laser welding, 2s of robot movement, 1s of idle state).

2.2 *Unit definition*

Several units have been evaluated to be used in the assessment. Existing literature on the topic commonly performs assessments per kg of matter processed (Dalquist 2004, Duflou 2012, Shah 2023, Le 2024), including beyond assessments specific to metal forming or to digital fabrication techniques. Therefore, to align with the state of the art and obtain comparable data, evaluating impacts per kg of matter processed is considered as first unit in this research. However, it is also common to perform the assessment for volumes of matter (m^3) (see for example Munch-Petersen & Beim 2022). Given the nature of the ASLM and the material scarcity it leverages, evaluating impacts per m^3 of matter not only allows for comparisons with existing data in the literature, it also is a more relevant unit to understand the process itself. This is therefore the second unit considered in the present paper.

Finally, while measuring impacts per rod of matter used is unusual in the literature and does not allow to establish direct comparisons with other processes assessed, it is the basic unit of ASLM. This makes it a relevant unit to examine, not for comparison to other forming processes, but to understand how to guide design principles based on environmental impacts within ASLM. Also because of this particularity of ASLM, some results are presented detailing the specific type of unit-cell considered in the lattice, as well as the number of rods and density that the use of such unit-cell entails. Given the high changeability of this parameter, a sensitivity study for the various type of unit-cells available is also performed on it.

2.3 *Phases and impacts assessed*

The research only considers energy costs, assessed over A1 (Raw material extraction) - A3 (Manufacturing) phases according to the ISO 14040-14044 framework. Preliminary results are also presented regarding greenhouse gas emissions associated to electricity use and to raw material production. The latter is achieved by relying on data found in prior literature.

3 RESULTS

3.1 *Per unit*

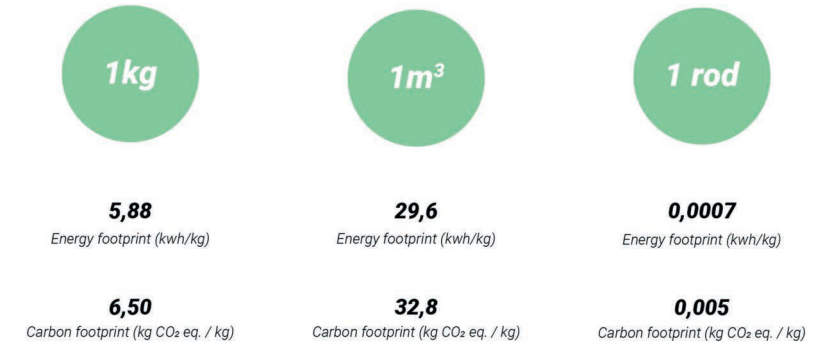


Figure 3. Energy footprint and carbon footprint sum-up per unit assessed.

Figure 3 sums up results for the three units assessed, considering a diamond lattice unit-cell (5856 rods or 5.04 kg). Results are detailed further in Table 1.

Table 1. Sum-up of results per unit assessed.

	Embedded energy (kWh/ unit)	Process energy (kWh/ unit)	Total energy cost (kWh/ unit)	Material GHG emissions (kg CO_2 eq./unit)	Process GHG emissions (kg CO_2 eq./ unit)	Total GHG emissions (kg CO_2 eq./ unit)
1 kg	5,60E+00	2,80E-01	**5,88E+00**	6,15E+00	3,50E-01	**6,50E+00**
1 m³	2,82E+01	1,41E+00	**2,96E+01**	3,10E+01	1,76E+00	**3,28E+01**
1 rod	4,82E-04	2,32E-04	**7,14E-04**	5,29E-03	1,39E-05	**5,31E-03**

3.2 *Energy footprint breakdown*

Figure 4 presents findings relative to the energy footprint of each part of the system. ASLM mobilizes the following equipment:

- ABB I1660ID;

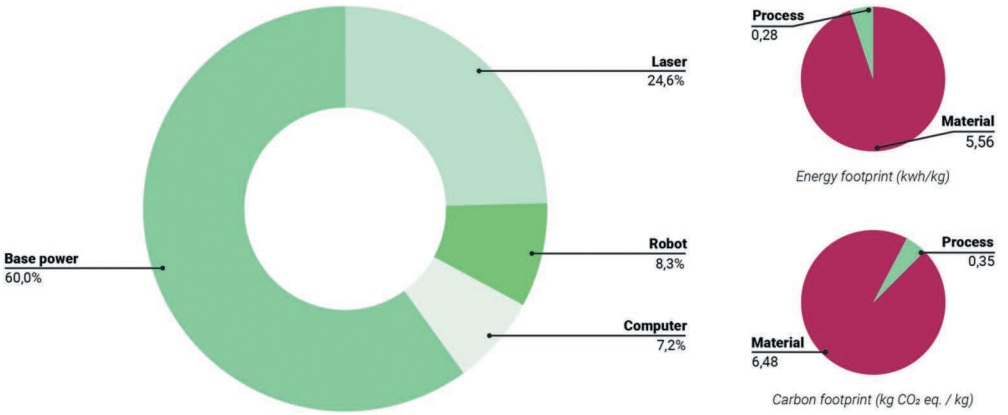


Figure 4. Energy footprint breakdown for the ASLM process.

- ABB IRC5 compact (single phase);
- MAXphotonics 1kW fiber laser;
- 2 PI E-617 piezo amplifiers;
- 4 Basler Dart cameras.

Table 2 presents the measured energy consumption on-site during the fabrication process, for the assembly of one rod into the lattice. This assembly has an average duration of 3,6 seconds (0.6s of laser welding, 2s of robot movement, 1s idle state). Furthermore, the average laser consumption if considering only power peaks is 534J per rod.

Figure 4 also presents the relationship between this measured process energy footprint and the embedded energy of the material itself. It shows the very low energy footprint of the process, as 95% of the energy costs in ASLM originates in the material itself. A similar relation between material and system costs exists in the partial carbon footprint assessment performed, with 95% of the impact originating in the material. Table 3 and 4 provide the detailed figures, considering 6,00E-02 kg CO_2 eq. / kWh for the French energy mix (Carbone4).

Table 2. Energy footprint breakdown per part of the system.

	Energy consumption (J)
Laser	205
Robot	69
Computer (Prieto 2022)	60
Base Power	500
Total	**834**

Table 3. Energy consumption breakdown per material and system.

Rod embedded energy (kWh/kg) (Teubler 2019)	5,56E+00
Process energy use (kWh/kg)	2,80E-01
Total (kWh/kg)	**5,84E+00**

Table 4. Carbon footprint breakdown per material and system.

Global Warming Potential - Rod (kg CO2 eq / kg) [1] (Carboncloud)	6,48E+00
Global Warming Potential - Process (kg CO2 eq / kg) [2]	3,50E-01
Total (kg CO2 eq / kg)	**6,83E+00**

4 COMPARISON WITH OTHER METAL FORMING PROCESSES

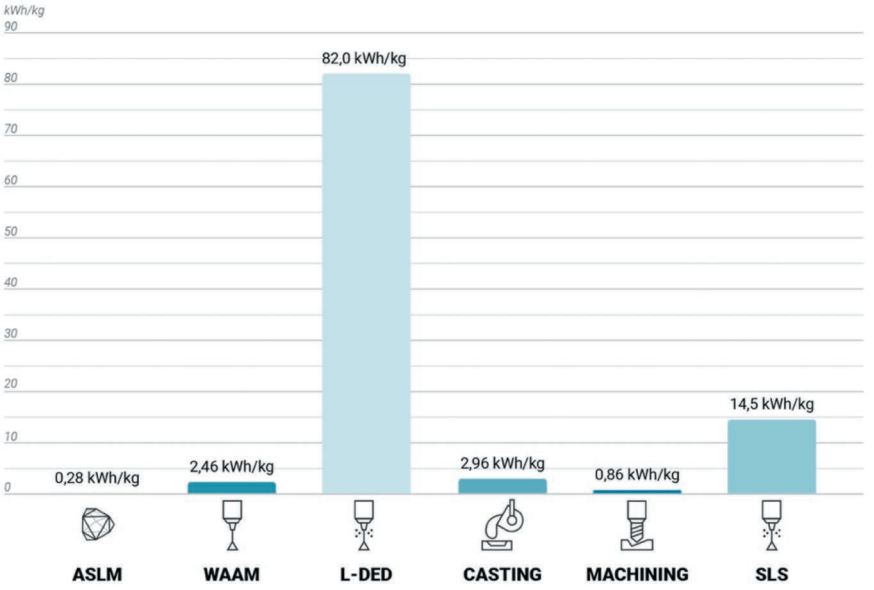


Figure 5. Energy consumption comparison for various metal forming processes.

Figure 5 presents a general comparison between ASLM and various other metal forming processes, with results detailed in Table 5. Table 6 presents a more detailed comparison between ASLM and laser powder-based fusion. For ASLM, a 316L SS diamond lattice unit-cell (5856 rods or 5.04 kg) with rods ø 1.6 x 55mm and 3.6 s per rod cycle is considered. For laser powder-based fusion, 50μm powders and build speed 0.5g/min is considered (Teubler 2019).

Table 5. Energy footprint comparison with various metal forming processes.

Process	Energy footprint for 1 kg (kWh)
ASLM	0,28
Wire Arc Additive Manufacturing (Shah 2023)	2,46
Laser Directed Energy Deposition (Shah 2023)	82
Casting (Le 2024)	2,96
Fine Machining (Dalquist 2004)	0,86
Selective Laser Sintering (Duflou 2012)	14,5

Table 6. ASLM and L-PBF comparison data.

	Material embodied energy (kWh/kg)	Manufacturing time for 1kg (h)	Manufacturing time for 1m^3 (h)	Process energy for 1kg (kWh)	Process energy for 1m^3 (kWh)	Total energy footprint for 1m^3 (kWh)
ASLM	5,6	1,20	6	0,28	1,42	**29,6**
L-PBF	23	33,00	166	276	1390	**1506**

4.1 *Unit-cell sensitivity analysis*

ASLM manufacturing allows for great flexibility in the type of lattice implemented. Various types of unit-cells have different densities which might affect the energy consumption per m^3 of the process. Figure 6 presents the sensitivity study performed to evaluate the impact of different unit-cell types on energy consumption, considering rods ø 0,8mm and length 55mm.

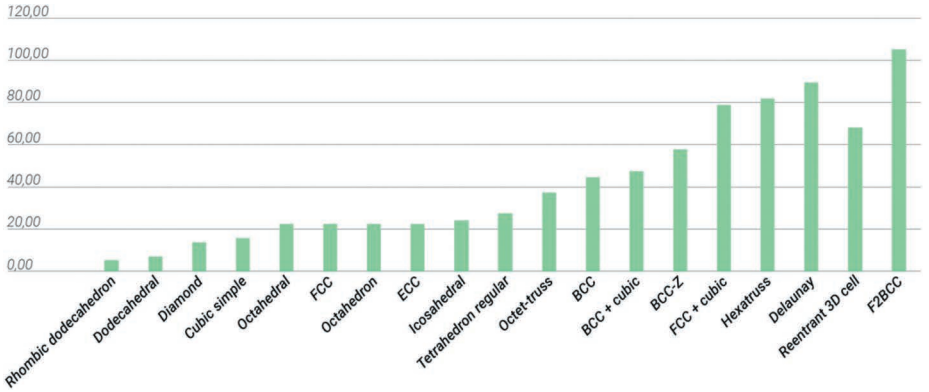


Figure 6. Sensitivity study for energy density per unit-cell type (J/cm^3).

Table 7 presents the detailed results of the energy density evaluation per unit-cell type, as well as the greenhouse gas emissions associated to each, considering both the emissions associated with the material and the emissions associated with the process energy. As the energy efficiency of the unit-cells correlates with these emissions, the best performing unit-cells are the same as highlighted in Figure 6.

Table 7. Energy and carbon footprint per unit-cell type.

Unit-cell	Energy density (J/cm^3)	Global Warming Potential (Kg CO2 eq./cm^3)[2]
Diamond	5,12	1,11E-06
Dodecahedral	6,86	1,48E-06
Rhombic dodecahedron	13,65	2,95E-06
Cubic simple	15,76	3,40E-06
Octahedral	22,29	4,81E-06
FCC	22,29	4,81E-06
ECC	22,29	4,81E-06
Octahedron	22,29	4,81E-06
Icosahedral	24,08	5,20E-06
BCC	27,30	5,90E-06
Octet-truss	37,15	8,02E-06
Tetrahedron regular	44,57	9,63E-06
BCC-Z	47,28	1,02E-05
BCC + cubic	57,79	1,25E-05
FCC + cubic	78,80	1,70E-05
Hexatruss	81,89	1,77E-05
Delaunay	89,30	1,93E-05
F2BCC	68,24	1,47E-05
Reentrant 3D cell	105,06	2,27E-05

5 DISCUSSION AND OUTLOOK

The present research place of ASLM within recent steel AM developments from an energy perspective, demonstrating the drastic reduction of consumption associated with the process and confirming the potential of ASLM for the AEC industry. Results show the very significant reduction of energy consumption in the ASLM process compared to other metal forming processes, from 67,5% to 99,7% of reduction depending on the technology it is compared with, and from 88,6% to 99,7% of reduction compared to other steel AM technologies.

Part of the present assessment however relies on a comparison considering kWh consumed per kg of material processes published in the literature. While assessments with identical

measurement methods have been privileged, discrepancies might exist in the system boundaries for each of the assessments relied on, which would request further evaluation to confirm results. Limits to this study also include the restriction to energy consumption. To assess impact transfers that are potentially at play in digital manufacturing processes for the AEC (Kuzmenko 2020), a complete LCA of ASLM is still necessary. Complementary studies should also be performed to consider the adaptability of ASLM for different material types, such as metals and thermoplastic polymers, a key advantage for the dissemination of the technique but a major factor of change in calculating its impacts.

Finally, not only is the ASLM process able to produce a large range of various truss configurations for AEC products, based on different lattice cells and with differentiated scales made possible, but the sensitivity study also shows that the type of unit-cell has a strong weight in the system as impacts can vary tenfold. This can be leveraged directly in design work with ASLM, transforming analytic LCA into a performative design tool. Case studies such as an I-beam with varying truss configurations, documenting the variation in energy consumption, would therefore open up new discussions on truss discretization for topological optimization in structural design.

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